

RSA DSA

Module V

RSA Algorithm

Public key generation

- **Steps:** generate public and private key, encrypt using public key and decrypt using the private key.
- **Rules for generating a public key:**
- Select two prime numbers p and q .
- Compute $n = p * q$
- Compute $\phi(n) = (p-1)*(q-1)$
- Select a number e such that e is $1 < e < \phi(p) * \phi(q)$ and $\gcd(e, \phi(p) * \phi(q)) = 1$.
- The public key is $\{ e, n \}$.
- Find a number d such that,
$$e * d \bmod \phi(n) = 1 \text{ (d is a multiplicative Inverse here)}$$

The private key is $\{ d, n \}$.

RSA-DSA

The hash code $H(M)$ is generated for the Input message M . Further, the hash code $H(M)$ is encrypted (signed) with the private key (PR_a) of a sender A to generate a signature s . The message m along with the signature s is forwarded to the receiver. At receiver side, the signature is verified using public key of PU_a sender by decrypting the signature value. The decrypted output is received hash code. Further, hash code is computed with the received message M . The computed hash code is compared with the received hash code for the verification of the signature. If the computed hash code and received hash code is same then the received message is original message with out any modifications happens during the transmission through the network, otherwise, receiver discards the message.

